

Nickel Exploration Potential in Ontario

Spatial analysis of mineral occurrences and favourable geological settings

A mineral prospectivity study based on open data from the Ontario Mineral Inventory, the bedrock geology of Ontario, QGIS and Python

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Tools: QGIS — Python — Ontario Mineral Inventory — GeologyOntario

1. Introduction

Ontario is one of Canada's major mining provinces. Its geological basement, which includes a significant part of the Canadian Shield, has supported intense mineral exploration and mining activity for more than a century, particularly in base metals and strategic metals.

Nickel occupies a particular place within this landscape. It is a strategic metal used in alloy manufacturing, in numerous industrial processes, and, more recently, in the production chain for electric vehicle batteries. This dual demand, both industrial and energy-related, makes it a metal that is closely monitored in critical resource analyses.

Identifying areas favourable to nickel exploration rarely relies on a single source of information. It requires combining geological, spatial and mining data to build a coherent reading of the territory. Open datasets published by provincial geological agencies now offer a real opportunity to carry out this type of analysis in a reproducible way, without relying on proprietary data.

This report builds on that logic. It does not aim to demonstrate the existence of exploitable deposits, nor to replace technical field studies. It proposes a first spatial ranking of Ontario's districts based on the signals available in open data: the location of nickel occurrences, the lithological context, and mining status.

Research question: how can open geological data be used to identify the districts most favourable to nickel exploration in Ontario?

The central working hypothesis of this study is that counting nickel occurrences alone is not enough to rank districts. Three complementary dimensions need to be combined:

- the location of nickel occurrences at the provincial scale;
- their mining status, which reflects their level of maturity;
- their association with rocks that are geologically favourable to nickel.

This combination is the guiding thread of the entire report.

2. Data, Tools and General Method

The study is based on two open datasets published by GeologyOntario, supplemented by reference mapping layers, and on a combined QGIS and Python workflow.

2.1 Ontario Mineral Inventory (OMI)

The Ontario Mineral Inventory lists all known mineral occurrences in the province, totalling 18,873 occurrences. The fields used in this study are the following:

- MDI_IDENT — unique identifier of the occurrence;
- STATUS — mining maturity level of the occurrence;
- NAME — name of the occurrence;
- P_COMMOD — primary commodity;
- S_COMMOD — secondary commodity(ies);
- RGP_DIST — regional geological district;
- LATITUDE / LONGITUDE — geographic coordinates.

2.2 Bedrock Geology of Ontario — MRD126-REV1

The bedrock geology of Ontario is documented by the Geopoly.shp layer, part of the Bedrock Geology of Ontario dataset (MRD126-REV1), which includes 20,956 geological polygons. The key field used is ROCKTYPE_P, which records the rock type associated with each polygon and allows favourable nickel lithologies to be identified: mafic and ultramafic rocks, gabbro, peridotite, pyroxenite and komatiite.

2.3 Complementary Layers

- administrative boundary of the province of Ontario;
- OpenStreetMap basemap;
- intermediate layers produced in QGIS;
- final enriched table, produced with Python.

2.4 General Approach

The method follows a sequential logic: extraction of nickel occurrences, identification of favourable rocks, spatial overlay between the two, construction of a prospectivity score by district, followed by verification and enrichment of this score using Python. Results are presented as tables and maps, each accompanied by an interpretation.

3. Extraction of Nickel Occurrences

Nickel occurrences were filtered in QGIS from the complete OMI database, querying both the primary and secondary commodity fields simultaneously:

```
"P_COMMOD" ILIKE '%Nickel%' OR "S_COMMOD" ILIKE '%Nickel%'
```

This filter extracted 1,554 nickel occurrences from the 18,873 occurrences in the complete OMI database.

It is important to stress that these 1,554 occurrences do not correspond to mines. They are mining objects with very different statuses: simple showings, discretionary occurrences, prospects, deposits, past producing mines, or producing mines. This heterogeneity is precisely what the rest of the analysis seeks to account for.

3.1 Breakdown by District

District	Nickel occurrences	Share
Timmins	375	24.1%
Sudbury	351	22.6%
Kirkland Lake	262	16.9%
Thunder Bay South	177	11.4%
Kenora	135	8.7%
Thunder Bay North	92	5.9%
Southern Ontario	64	4.1%
Red Lake	53	3.4%
Sault Ste. Marie	45	2.9%
Total	1,554	100%

Source: Ontario Mineral Inventory. Processing: QGIS.

In raw numbers, Timmins ranks first with 375 occurrences, ahead of Sudbury and Kirkland Lake. This initial ranking, however, remains purely quantitative: it says nothing about the maturity of the occurrences, nor about their consistency with a favourable geological context. It is only a descriptive starting point.

4. Analysis by Mining Status

The OMI STATUS field allows the 1,554 nickel occurrences to be broken down according to their level of mining maturity.

Status	Occurrences	Share
Occurrence	912	58.7%
Discretionary Occurrence	285	18.3%
Prospect	120	7.7%
Past Producing Mine	101	6.5%
Deposit	73	4.7%
Past Producing Mine (Low Tonnage)	54	3.5%
Producing Mine	9	0.6%

Source: Ontario Mineral Inventory. Processing: QGIS.

The majority of nickel occurrences correspond to simple occurrences or discretionary occurrences, which together account for more than three-quarters of the total. A reading based solely on point counts can therefore overstate districts where indications are numerous but not very advanced.

Conversely, districts with fewer occurrences may prove more interesting if those occurrences correspond to prospects, deposits or past producing mines. This is the logic behind the construction, later in the report, of a score weighted by mining status rather than a simple count.

5. Overlay with Mafic and Ultramafic Rocks

5.1 Extraction of Favourable Rocks

Favourable nickel lithologies were identified from the ROCKTYPE_P field of the geological layer, by combining several rock categories associated with nickel-bearing mineralization:

```
"ROCKTYPE_P" ILIKE '%mafic%' OR "ROCKTYPE_P" ILIKE '%ultramafic%' OR  
"ROCKTYPE_P" ILIKE '%gabbro%' OR "ROCKTYPE_P" ILIKE '%peridotite%' OR  
"ROCKTYPE_P" ILIKE '%pyroxenite%' OR "ROCKTYPE_P" ILIKE '%komatiite%'
```

This expression isolated 5,087 mafic or ultramafic geological units within the 20,956 polygons of the Bedrock Geology of Ontario layer. These units represent lithological contexts favourable to Ni-Cu-PGE-type mineralization. They do not prove the presence of nickel, but they make it possible to filter nickel occurrences according to their geological compatibility.

5.2 Spatial Overlay

The spatial overlay was carried out in QGIS between the layer of 1,554 nickel occurrences (OMI_nickel_1554) and the layer of mafic and ultramafic units (Geopoly_mafic_ultramafic), using the geometric predicate “intersects”.

This overlay identified 589 nickel occurrences located on mafic or ultramafic rocks, representing 37.9% of the 1,554 initial nickel occurrences.

District	Occurrences on favourable rocks
Sudbury	162
Timmins	141
Kirkland Lake	100
Thunder Bay South	59
Thunder Bay North	34
Red Lake	32
Kenora	31
Southern Ontario	17
Sault Ste. Marie	13
Total	589

Source: Ontario Mineral Inventory; GeologyOntario MRD126-REV1. Processing: QGIS.

The lithological filter significantly changes the ranking observed in Section 3. Before the overlay, Timmins ranked first in raw volume. After overlaying with mafic and ultramafic rocks, Sudbury moves to first place with 162 favourable occurrences, ahead of Timmins and Kirkland Lake. This result shows that exploration potential does not depend solely on the density of occurrences, but also on their location within a favourable geological context.

6. Building the Prospectivity Score

To account simultaneously for the density of favourable occurrences and their mining maturity, a prospectivity score was built for each district, based on a scale assigning an increasing weight to each mining status.

Status	Score
Occurrence	1
Discretionary Occurrence	1
Prospect	2
Deposit	3
Past Producing Mine	4
Past Producing Mine (Low Tonnage)	4

Status	Score
Producing Mine	5

Source: Ontario Mineral Inventory. Processing: QGIS.

The final score is calculated only on the 589 nickel occurrences located on favourable rocks, by summing for each district the weights corresponding to the status of each of its occurrences.

Rank	District	Favourable occ.	Final score	Class
1	Sudbury	162	278	Very strong
2	Kirkland Lake	100	220	Strong
3	Timmins	141	193	Strong
4	Thunder Bay South	59	74	Intermediate
5	Thunder Bay North	34	45	Intermediate
6	Kenora	31	36	Moderate
7	Red Lake	32	35	Moderate
8	Southern Ontario	17	22	Low
9	Sault Ste. Marie	13	16	Low

Source: Ontario Mineral Inventory; GeologyOntario MRD126-REV1. Processing: QGIS.

Sudbury emerges as the most robust district in the analysis. It combines a high number of occurrences on favourable rocks with generally advanced mining statuses. Kirkland Lake overtakes Timmins despite a smaller number of favourable occurrences, because its average profile is more mature. Timmins remains an important district, but its score is driven more by the volume of occurrences than by their average maturity.

7. Complementary Python Analysis

A second processing step was carried out in Python, in order to verify and make reproducible the score calculation performed in QGIS. For environment-related reasons (dependency management via uv), the script was written using only the Python standard library, without relying on pandas or matplotlib.

Three additional indicators were added at this stage:

- the district's share of total favourable occurrences;
- the district's share of the total score;
- a maturity index, defined as the final score divided by the number of favourable occurrences.

The table below presents the enriched result, exported from QGIS and then recalculated and completed in Python:

Rank	District	Occ.	Score	Class	Occ. %	Score %	Maturity
1	Sudbury	162	278	Very strong	27.5	30.3	1.72
2	Kirkland Lake	100	220	Strong	17.0	23.9	2.20
3	Timmins	141	193	Strong	23.9	21.0	1.37
4	Thunder Bay South	59	74	Intermediate	10.0	8.1	1.25
5	Thunder Bay North	34	45	Intermediate	5.8	4.9	1.32
6	Kenora	31	36	Moderate	5.3	3.9	1.16
7	Red Lake	32	35	Moderate	5.4	3.8	1.09
8	Southern Ontario	17	22	Low	2.9	2.4	1.29
9	Sault Ste. Marie	13	16	Low	2.2	1.7	1.23

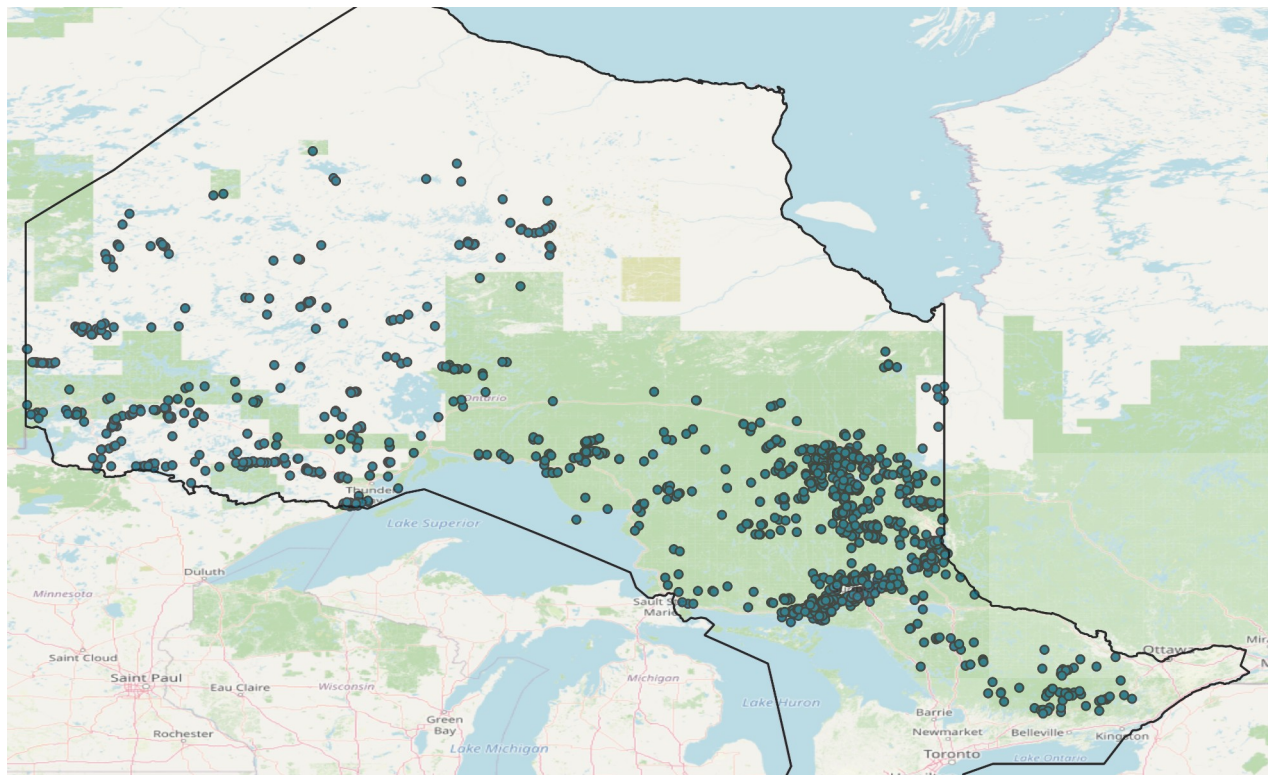
Source: Ontario Mineral Inventory; GeologyOntario MRD126-REV1. Processing: QGIS / Python.

Sudbury accounts for 27.5% of favourable occurrences, but 30.3% of the total score: the district therefore combines both density and maturity. Kirkland Lake represents only 17.0% of favourable occurrences, but 23.9% of the total score, with the highest maturity index in the study (2.20), meaning that its occurrences are on average more advanced. Timmins accounts for 23.9% of favourable occurrences, but only 21.0% of the total score, with a lower maturity index (1.37): the district remains important in volume, but less mature on average.

The contribution of Python at this stage is twofold. On one hand, it makes the ranking reproducible from the table exported from QGIS. On the other hand, it goes beyond the simple final score by distinguishing density from average maturity. This step limits manual errors and strengthens the credibility of the approach within a portfolio logic.

8. Mapping and Interpretation of the Maps

8.1 Map 1 — General Distribution of Nickel Occurrences in Ontario



Map 1 — General distribution of nickel occurrences in Ontario

Source: Ontario Mineral Inventory; Ontario GeoHub; OpenStreetMap. Processing: QGIS.

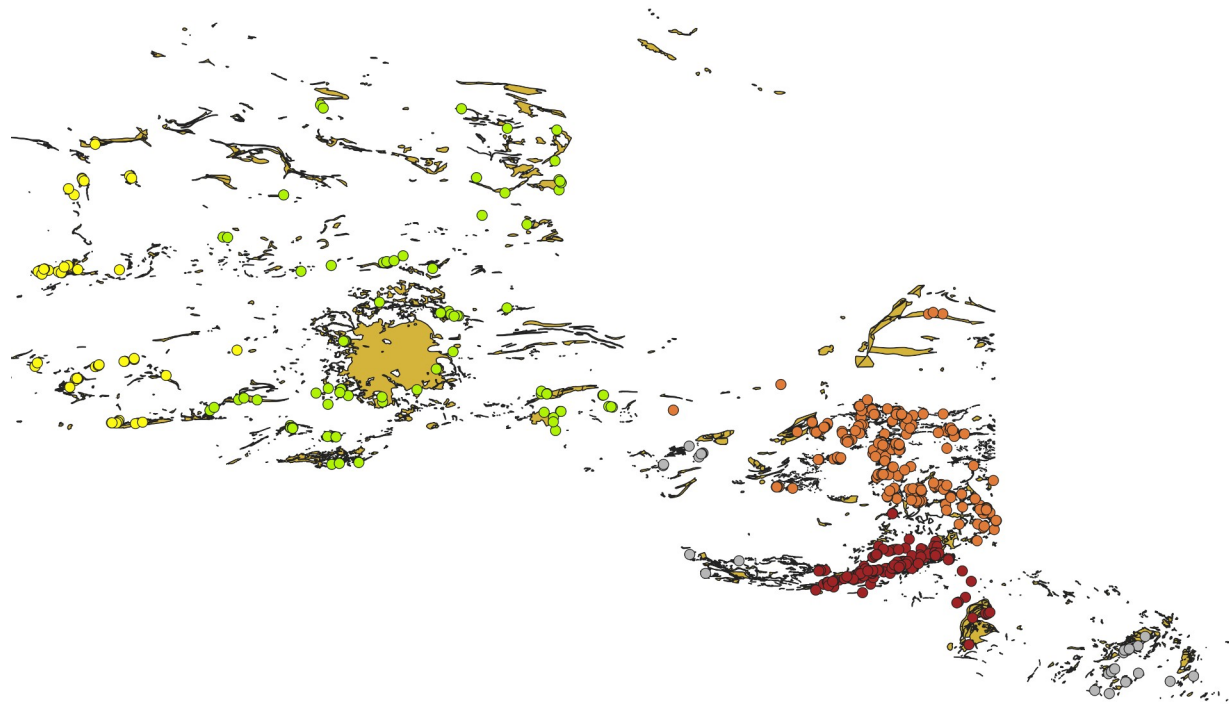
This map serves as the context map for the study. It shows the 1,554 nickel occurrences extracted from the Ontario Mineral Inventory at the provincial scale. Each point corresponds to an occurrence where nickel is recorded as a primary or secondary commodity. These points should not be interpreted as mines: they may correspond to simple showings, prospects, deposits, past producing mines, or active mines.

The map allows the raw distribution of occurrences to be visualized before any geological filtering and before any weighting by mining status. It highlights a strong concentration in northern and northeastern Ontario, particularly around the districts of Timmins, Sudbury and Kirkland Lake. It also shows more scattered occurrences in the districts of Thunder Bay, Kenora, Red Lake, Southern Ontario and Sault Ste. Marie.

This map does not rank exploration potential. It does not distinguish between mining statuses

and provides no indication of nickel grades or the economic value of the occurrences. It serves only as a spatial starting point, showing where known nickel occurrences are located within the OMI database.

8.2 Map 2 — Nickel Exploration Potential in Ontario



Map 2 — Nickel exploration potential in Ontario

Source: Ontario Mineral Inventory; GeologyOntario MRD126-REV1; Ontario GeoHub. Processing: QGIS / Python.

The basemap was deliberately simplified in order to better highlight the potential classes and the favourable rocks, which are the main focus of this map.

This map constitutes the final potential map. Unlike the previous one, it does not show the 1,554 initial nickel occurrences, but only the 589 occurrences located on mafic or ultramafic rocks. This lithological filter makes it possible to focus the analysis on the geological contexts most favourable to Ni-Cu-PGE-type mineralization.

The colours correspond to the potential classes calculated for each district from the final score. They do not represent nickel grades, resources, or economic value.

- dark red: very strong potential — Sudbury;
- orange: strong potential — Kirkland Lake and Timmins;

- yellow / light green: intermediate potential — Thunder Bay South and Thunder Bay North;
- grey / pale green: moderate or low potential — Kenora, Red Lake, Southern Ontario and Sault Ste. Marie.

The map allows a shift from simple location to spatial prioritization. Sudbury stands out as the main core of potential, as it combines occurrence density, a favourable lithological context and mining maturity. Kirkland Lake and Timmins also appear as two strong hubs, but with different profiles: Kirkland Lake is less dense but more mature, while Timmins is very dense but less advanced on average. The Thunder Bay districts appear as intermediate zones, less of a priority than the top three but not negligible.

This map should be read as a map of relative prospectivity, not as a map of mineral wealth. It is intended to help identify the districts to examine first within an exploratory approach.

9. General Discussion

The most important result of this study is the progressive shift in district ranking as successive criteria are added to the analysis:

- in raw volume, Timmins ranks first with 375 nickel occurrences;
- after the lithological filter, Sudbury moves ahead with 162 occurrences on favourable rocks;
- after weighting by mining status, Sudbury remains first with a final score of 278;
- Kirkland Lake overtakes Timmins thanks to a higher maturity index;
- Timmins remains important in volume, but its average profile is less advanced;
- the Thunder Bay districts form secondary but interesting zones;
- the districts of Kenora, Red Lake, Southern Ontario and Sault Ste. Marie show more limited potential according to this method.

Prospectivity cannot be reduced to a density of occurrences. A district may contain many indications without being a priority if those indications are not very mature or are poorly associated with favourable lithologies. Conversely, a less dense district may stand out strongly if its occurrences show more advanced statuses.

10. Limitations

This study relies exclusively on open data and a simplified method. Several limitations should be highlighted:

- absence of consistent nickel grade data;
- absence of tonnage data;
- absence of systematic resource and reserve data;
- dependence on the OMI STATUS field, which may be unevenly updated across occurrences;
- lithological selection performed using keywords, which may omit certain naming variants;
- variable precision of the geological layers depending on the area of the province;
- absence of geophysical data;
- absence of distance to roads, towns, ports, railways or electrical infrastructure;
- deliberately simplified prospectivity score;
- absence of field validation.

This study does not aim to estimate the real economic value of potential deposits. It ranks relative spatial prospectivity, not metal wealth. The score should be understood as an initial prioritization tool, allowing districts to be identified for closer examination, rather than as an economic valuation model.

11. Conclusion

This study shows that nickel exploration potential in Ontario does not depend solely on the number of recorded occurrences. The overlay between Ontario Mineral Inventory occurrences, mafic and ultramafic rocks, and mining status identifies Sudbury as the most robust district, ahead of Kirkland Lake and Timmins.

Sudbury combines density, maturity and geological consistency. Kirkland Lake stands out with the highest maturity index in the analysis, while Timmins retains an important role due to its volume of occurrences, but with a less advanced average profile.

The combined use of QGIS and Python makes it possible to build a first reproducible ranking of nickel potential from open data. This approach remains exploratory and could be enriched by adding grade, tonnage, resource, infrastructure, geophysical and field validation data.

12. Data Sources

- Ontario Mineral Inventory — [GeologyOntario](#)
- Bedrock Geology of Ontario, MRD126-REV1 — [GeologyOntario](#)
- Ontario GeoHub — [administrative boundaries](#)
- OpenStreetMap — [basemap](#)
- Personal processing: QGIS and Python, 2026